



Technical Service Bulletin: TSB220058

Released Date: 30-Mar-2022

Valve Guide Wear Caused by Oil and Carbon Buildup from Valve Stem Seal: Black Smoke, Engine Noise, or Lower Power

Valve Guide Wear Caused by Oil and Carbon Buildup from Valve Stem Seal: Black Smoke, Engine Noise, or Lower Power

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QST30 CM552
- QST30 PG CM850

Issue Summary

Symptoms:

- Excessive black smoke
- Abnormal engine noise
- Low power

Root Cause:

- Existing Tier 2 valve stem seal is **not** intended for the engine operating exhaust port pressures and hence, does **not** seal the exhaust gases or meter oil per requirement.

Verification

- Upon inspection would oil/soot mixture found packed in the cylinder head near valve stems and springs and valve stem seals.
- Upon removal of the valve stem guides and valves severe amount of corrosion, carbon deposits and material loss from valve guides is observed.

Resolution

Replace Tier 2 Valve Stem Seal with Tier 4 Valve Stem Seal and/or Replace entire cylinder head assembly with new part number.

1. Remove Suspect Cylinder Head from Engine

- See corresponding Service Manual. Reference Procedure 002-004 in Section 2.

2. Suspect Cylinder Head Removal Preparation

- Purchase a full set of QST30 cylinder heads. See Parts list in Appendix A1.1.
- Purchase all new parts as specified in parts list and reference reuse list for reworking used good cylinder heads in Appendix A1.2

3. Disassemble Suspect Cylinder Head

- Disassemble cylinder head following Procedure 002-004 and 002-020 in Section 2
- Skip removing exhaust and intake valve guides unless reworking good used cylinder head.
- Remove tier 2 valve stem oil seals.
- Skip Clean, Inspect, Leak and Pressure Test sections in 002-004 in Section 2 unless reusing good used cylinder head continue following all procedures.

4. Installing Replacement Cylinder Head

- Install replacement cylinder head. See corresponding Service Manual. Reference Procedure 002-004 in Section 2

5. Reworking Suspect Tier 2 Cylinder Head

Assembly Preparation

Gather all supplies and tools needed for performing assembly.

Cummins® Tools Needed Prior to Assembly:

- Valve guide driver, Part Number 6304260
- Valve seal driver, Part Number 3163429
- Valve spring compressor, Part Number ST-448 or Part Number 3163639, or equivalent
- Valve spring stand, Part Number ST-1022
- Compressor plate, Part Number 3375043

Assembly:

- First few cylinder heads assembled, measure all valve guide lengths, and record the value and compare with guide overall length dimensions located in Appendix B1.
- Install exhaust and intake valve guides using tool when reworking good used cylinder head
- Verify guide installs properly to 21.5 mm by measuring guide exposed difference (see B1.2).
- Install Tier 4 valve stem seals on all exhaust and intake valve guides by using tool, Part Number 3163429.
- Continue assembling cylinder head. See corresponding Service Manual. Reference Procedure 002-004 and 002-020 in Section 2.

Reason for Change

Cylinder head assembly was changed to counter recurring valve stem seal malfunctions in the field.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Cylinder Head	40686731	No	No	55936202
¹ All parts have already been installed with new (Tier 4) valve stem seal and valve stem guide pressed to proper depth. ² New part number in the process of being released.				

Service Tools and Training Available

New tool for installing valve stem guide to the proper depth has been released and will be in stock shortly. See Appendix B1 for dimensional print.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information		
ESN First	Build Date ¹	Plant
37284774	1 June 2021	Seymour Engine Plant
¹ Engine build date can be found on engine dataplate.		

Appendix A1

A1.1 - New Cylinder Head Parts List

Table 3, New Cylinder Head Parts List				
Item Number	Description	Part Number	Head Quantity	Minimum Purchase Quantity
13	Tier (T) 2 Cylinder Head Assembly	5593620	1	>12

Rework Parts List

Note : Based on Condition denotes following clean, inspection, and leak test procedures found on Procedure 002-004 and 002-020 in Section 2.

Reference assembly layout continued in Appendix A2 for item number references.

A1.2 - Rework Parts List

Table 4, Rework Parts List					
Item Number in Figure 1	Description	Part Number	Head Quantity	Reuse/Replace	Comments
13	T2 Cylinder Head Assembly	4068673	1	Reuse	Based on condition
-	T2 Cylinder Head Rebuild Kit	5579210	1	Purchase	None
-	T4 Valve Stem Seal	2897022	4	Purchase	None
8	Valve Collets	3092344	8	Replace	Discard and use new collets
9	Exhaust Valve Kit -1- Valve, 2- Collets	3804867	2	Replace - Valve stems and collets	Use new valve stems and collets
7	Intake Valve Kit -1-Valve, 2-Collets	3804866	2	Replace - Valve stems and collets	Use new valve stems and collets
6	T2 Valve Stem Seal	3092608	4	Replace	Discard and use T4 seals
14	Valve Stem Guide	3092014	4	Replace	Use new valve guides
17	Intake Valve Seat (Valve Insert)	3094159	2	Reuse	Based on condition
12	Exhaust Valve Seat (Valve Insert)	3092461	2	Reuse	Based on condition
4	Valve Spring Retainer	3092510	4	Reuse	Based on condition
2	Large Spring	3092508	4	Reuse	Based on condition
3	Small Spring	3092509	4	Reuse	Based on condition
5	Valve Spring Guide	3092511	4	Reuse	Based on condition

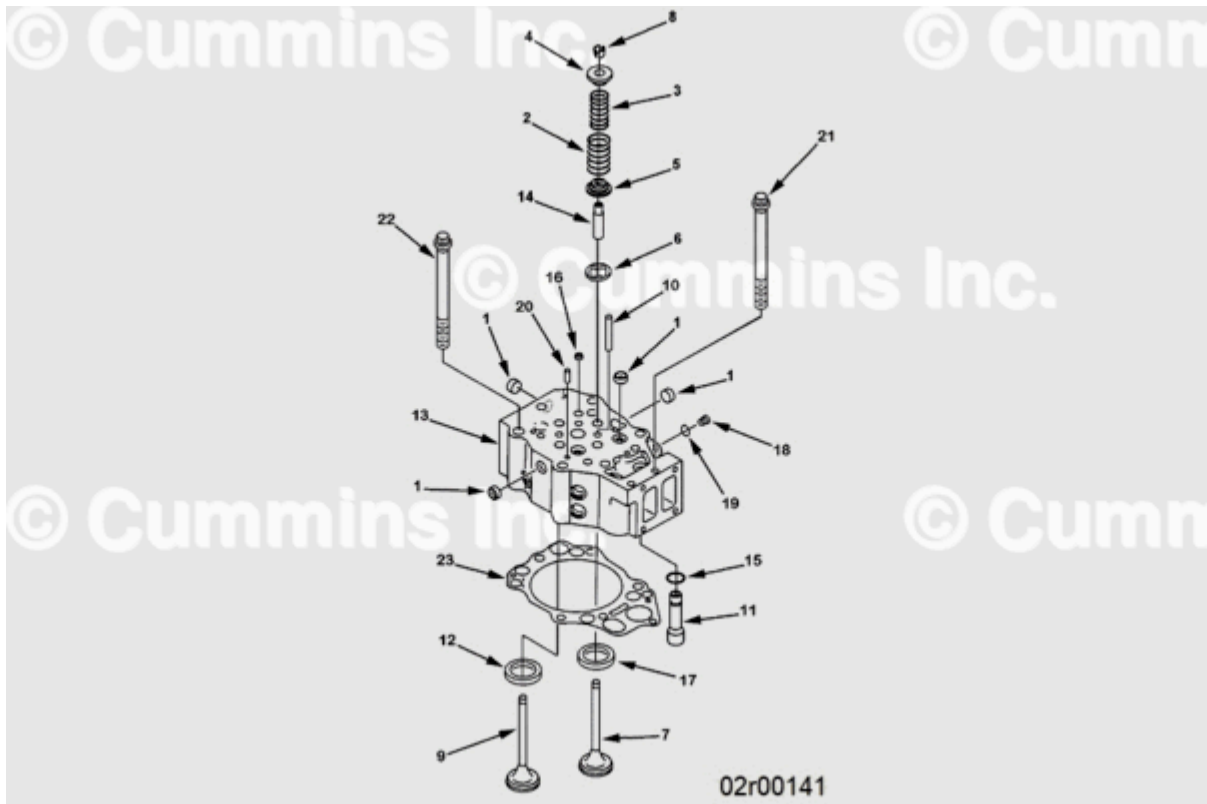


Figure 1, QST30 CM552 Cylinder Head Layout.

Appendix B1

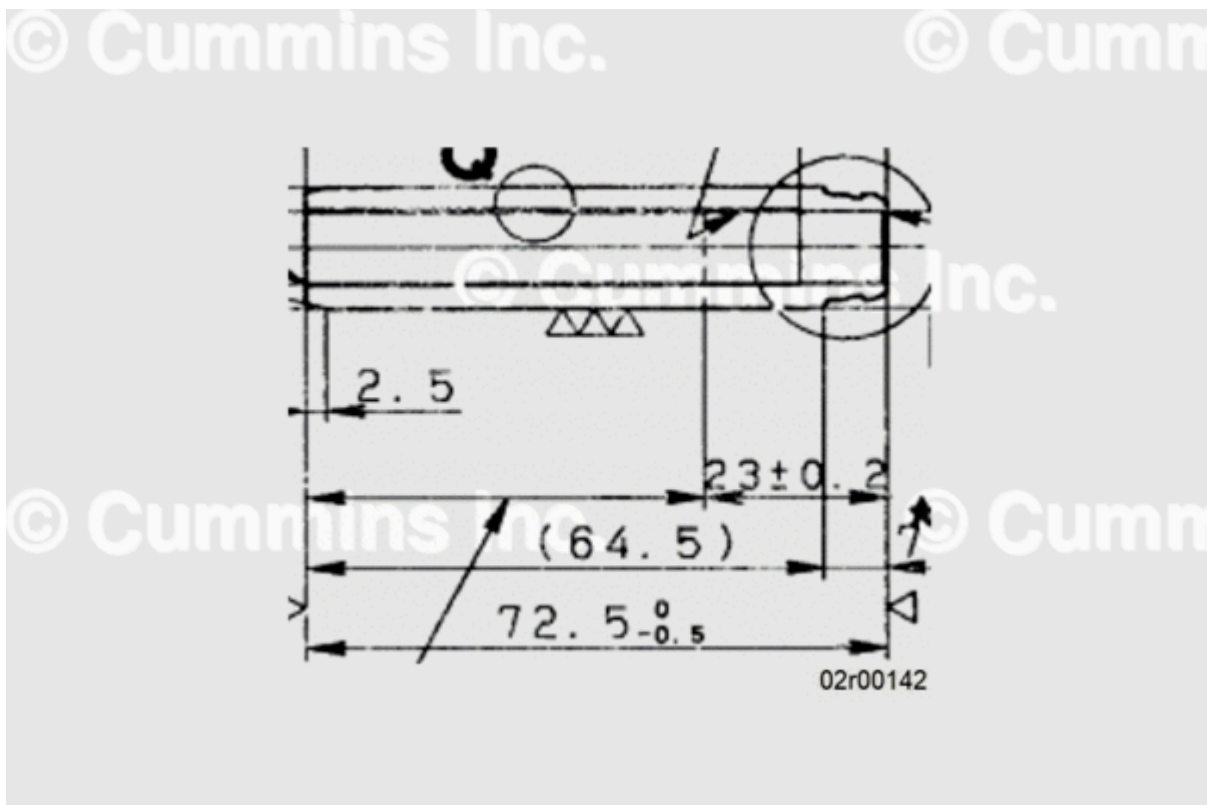


Figure 2, Valve Guide OAL: 72.5 0/-0.5 mm.

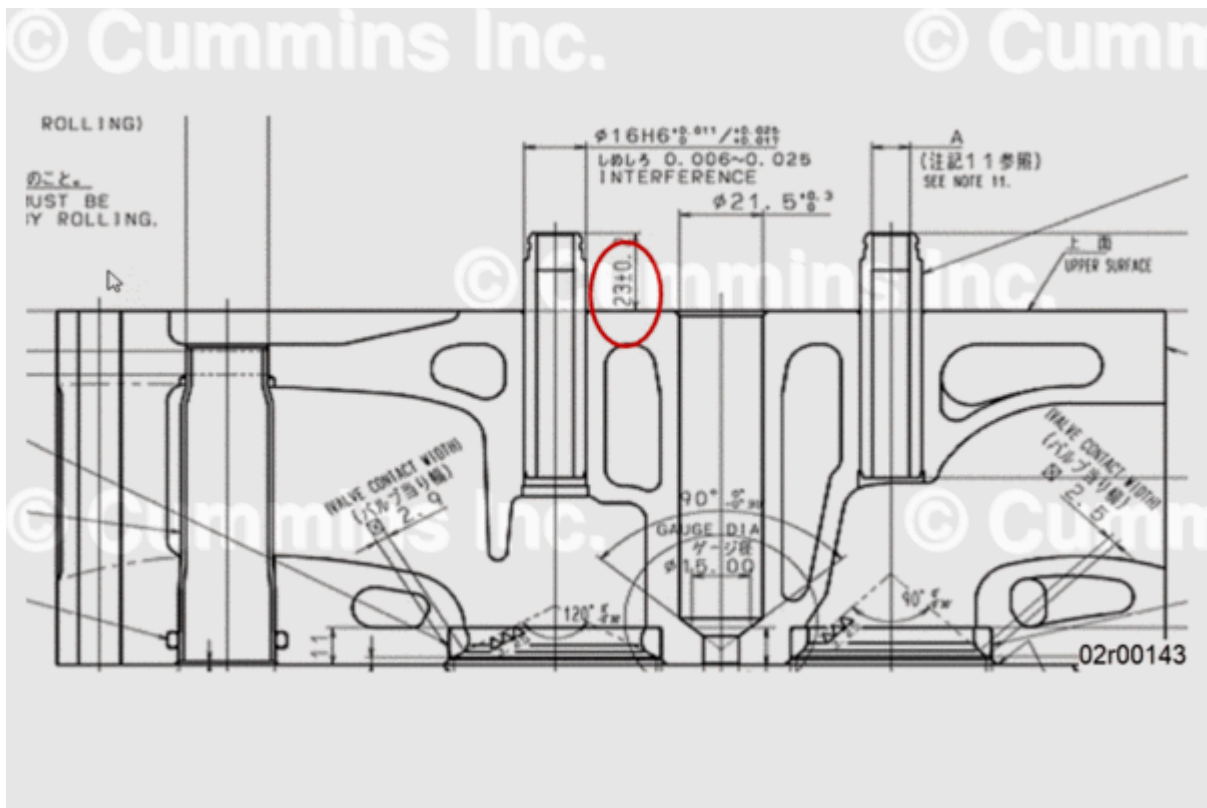


Figure 3, Valve Guide Install Depth Difference.

Previous Valve Guide Install Depth Difference 23 mm +/-0.2 mm

New Valve Guide Install Depth Difference 21.5 mm +/-0.2 mm

Document History

Date	Details
2022-3-16	Module Created

Last Modified: 30-Mar-2022